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Home: Arad (Romania)

ABOUT ME

I am a software developer with background in backend development and databases. I believe that almost any problem in our field can be solved if we invest enough time and effort to truly understand and address it.

My main skill is problem solving and systems thinking.

WORK EXPERIENCE

Accenture – Cluj-Napoca, Romania

Name of unit or department: AIP+

Software Developer

[01/07/2025 – Current]

Position as an AI Systems Software Developer/Engineer, with the main language being Python.

I've delivered critical Agentic builds for NATO and US state-level sales, facilitating integration and "activation" cycles

During this position, my main responsibilities were:

- Software development
- Feature Research
- Bug and incident investigation
- Bug fixing
- Updating documentation according to changes in the software. I managed the Release Note feature, which automatically generates information based on tasks from Azure DevOps
- Code review for colleagues working on the same project
- Discussions with internal clients
- Azure DevOps Pipeline Changes
- Kubernetes Deployments

This is a GenAI project focused on AI Agents.

Software Developer

[08/2021 – 30/06/2025]

Position as an Software Developer/Engineer, with the main language being C# on a .Net Project.

One achievement on this project was that I optimized 1TB data migration from 1 year to 2 weeks via parallel processing across 40 nodes for client retention.

During this position, my responsibilities included:

- Software development
- Bug and incident investigation
- Bug fixing
- Updating documentation according to changes in the software
- Code review for colleagues working on the same project
- Client discussions
- I was responsible for the DMT (Data Migration Tool) feature, used for migrating entities from legacy applications to our application
- Additionally, on this project, I have also acted as coordinator (mentor) for the internship program for two consecutive years, having mentored three interns so far
- I have also participated in and conducted technical interviews with potential interns for hiring in the third year

Programming languages used: Microsoft SQL, C#, Visual Basic, JavaScript, Python

 **Chamber of Commerce, Industry and Agriculture Arad** – Arad, Romania

IT Consultant

[01/01/2023 – 31/12/2024]

I analyzed and resolved a wide range of technical situations, providing solutions for the issues that arose.

Additionally, in the last 4 months, I focused on an end-to-end project:

I rebuilt and modernized the institution's website, starting from the old structure and using WordPress as the CMS for ease of management by internal staff.

Updated the design, completely restructured the pages, implemented with Astra and Elementor for flexibility and a professional look.

Custom development:

- Integrated JavaScript code for advanced filtering of accordion tables in course pages (tag filters: New, Free, All)
- Developed a custom plugin for dynamic search in Excel files, returning relevant information (e.g., Company Rankings), extendable for other scenarios
- Direct integration with Elementor, allowing the plugin to be added to any page

Security and access:

- Configured and customized Wordfence to prevent attacks and spam
- Implemented Prevent Direct Access to restrict access to Excel files used as internal databases
- Additional management at the .htaccess level and resolved PHP/WordPress errors

Optimization and functionality:

- Installed WP Statistics to monitor website traffic
- Relevanssi for improved internal search functionality
- Advanced Custom Fields for custom content display, especially for European projects

Technical support and data management:

- Uploaded databases via cPanel, synchronized between the local environment (XAMPP) and the server
- Troubleshoot and resolved plugin conflicts, compatibility errors, and data migration issues

 **Chamber of Commerce, Industry and Agriculture Arad** – Arad, Romania

Address: Strada Cloșca nr. 5, 310017 Arad (Romania) | **Website:** <https://www.ccia-arad.ro/> | **Business or sector:** Information and communication

Trainer

[01/04/2023 – 31/12/2024]

During my activity at the Chamber of Commerce, Industry and Agriculture Arad, I also had the opportunity to serve as a lecturer or trainer, delivering various courses and workshops in the IT field. More details can be found in the Conferences and Seminars section.

 **Continental Automotive** – Sibiu, Romania

Name of unit or department: MES - Manufacturing execution system - **Business or sector:** Manufacturing

Software Developer

[07/2018 – 06/2021]

Position more specialized as a MES Engineer, which meant I was a Software Developer/Engineer with the following activities:

- Production support for MES applications
- Creating reports using SQL, based on Oracle 12c
- Fixing and improving current applications in ANSI C or C#
- Developing new applications using ANSI C or C#
- Creating internal tools
- Organizing meetings
- Discussions with suppliers
- Conducting training sessions

Programming languages used: Oracle SQL, ANSI C, C#

Extra: Database creation, ETLs, participation in proof-of-concept projects.

 **Saguaro** – Timisoara, Romania

Business or sector: Professional, scientific and technical activities

Junior Software Developer

[04/2016 – 07/2017]

The main activity within this company was implementing tasks assigned by the team leader using the C++ programming language.

I've learned about version control and parallelism during this job.

 **Happy Tour** – Marbella, Spain

Business or sector: Accommodation and food service activities

Customer Support

[01/2014 – 08/2015]

My main activity consisted of:

Responding to emails from clients who requested more information about booking properties for vacations.

Additionally, in this role I also:

- Added the company's properties on various vacation websites for advertising
- Added, modified, and deleted various pieces of information on the company's website
- Sent banking details to clients to ensure payments were made and that the rented properties would not be double-booked

I made sure that, one week before each client's arrival, the staff in Marbella were informed to begin cleaning procedures and prepare to welcome the guests.

This job mainly involved IT, tourism, and administration.

I've learned time management and improved my communication skills

EDUCATION AND TRAINING

Trainer

Chamber of Commerce and Industry Neamț with Chamber of Commerce, Industry and Agriculture Arad [26/01/2023 – 27/03/2023]

Address: Strada Cloșca nr. 5, 310017 Arad (Romania) | **Website:** <https://www.ccia-arad.ro/> | **Field(s) of study:** Formarea Adulților

Bachelor's Degree – Software Developer

Universitatea de Vest 'Vasile Goldiș', Facultatea de Informatică [10/2010 – 07/2013]

Address: Bd. Revoluției nr. 85-87, Arad (Romania) | **Website:** www.uvvg.ro | **Field(s) of study:** Information and Communication Technologies

Principalele subiecte pe care le-am acoperit au fost:

1. Dezvoltare a sistemelor software folosind limbajele de programare: C++, Java, C#
2. Dezvoltare de pagini web folosind: HTML și PHP
3. Dezvoltare și interogare a bazelor de date folosind: MySQL

În această perioadă am mai fost plecat 3 luni prin programul Erasmus în Cehia la Universitatea Tomas Bata din Zlín.

High School Diploma

High School 'Vasile Goldiș' Arad [07/2006 – 06/2010]

Address: Calea Victoriei, Nr. 1-3, Arad (Romania) | **Website:** www.cnvga.ro | **Field(s) of study:** Information and Communication Technologies

LANGUAGE SKILLS

Mother tongue(s): Romanian

Other language(s):

English

LISTENING C2 READING C2 WRITING C2

SPOKEN PRODUCTION C2 SPOKEN INTERACTION C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Microsoft Office / Microsoft Excel / Adobe Photoshop

DRIVING LICENCE

Driving Licence: B

CONFERENCES AND SEMINARS

[14/07/2025 – 21/07/2025] Arad

Workshop: AI - Inteligența artificială I organized and delivered an interactive workshop dedicated to the use of artificial intelligence for facilitating project management and the drafting of project proposals. Main topics covered:

- Introduction to AI, with a focus on generative models
- Language models (LLMs), applications, and current trends
- Software agents and agent-based applications (such as GenSpark)
- Advanced prompting techniques and process automation (RAG, document creation)
- Practical use of ChatGPT, Gemini, and other AI models for content generation and digital workflows

The workshop was appreciated for its practical approach, interactivity, and adaptation to the group's needs, reflected in participant feedback, such as: "The lecturer is very well prepared, with genuine pedagogical skills. The course was presented in an interactive, concise manner and tailored to our needs." - participant feedback, Local Action Group

[24/09/2024 – 17/10/2024] Arad

Course - Competențe Digitale Trainer for the Digital Skills Course organized by the Chamber of Commerce, Industry and Agriculture Arad. During this course:

- We covered a brief history of the IT field
- We went through basic IT concepts
- We explored the applications included in the Office suite, with a focus on Excel
- Practical exercises with the students

The participants were aged between 20 and 50 years.

[18/08/2024 – 03/10/2024] Arad

Course - Web Design I designed and delivered a beginner web design course, structured around the fundamental steps of creating a modern website. I introduced basic concepts about website structure (HTML, CSS, containers, "inspect element"), followed by individual hands-on projects using HTML and CSS files. Participants learned how to integrate JavaScript for interactive features and explored concepts related to web communication and page requests. I encouraged the use of artificial intelligence for code generation, emphasizing the understanding of underlying logic and the development of autonomy in problem-solving. Later, the course included dedicated modules on modern platforms:

- Shopify
- WordPress (Local and Web)

Focusing on practice and adapting to the needs of non-technical participants, the course covered the design and launch of a website using current tools. The participants were between 20 and 60 years old.

[28/05/2024 – 18/06/2024] Arad

Course - Utilizarea Inteligentei Artificiale I designed and delivered an interactive course focused on artificial intelligence, providing both practical and theoretical insights for interested participants in Arad (ages 20–60). I

covered topics such as advanced language models (LLMs), generative AI, digitalization, and automation. Participants explored fundamental algorithms, Transformer architecture, and concrete applications:

- text generation (ChatGPT, Claude, Meta/Nvidia)
- image generation (Automatic1111 in Cloud, Leonardo)
- audio generation (Suno, ElevenLabs)

We also discussed the impact of AI on business processes and digital innovation, with hands-on examples and interactive Q&A sessions.

[22/04/2024 – 23/04/2024] Arad

Workshop: AI - Inteligența artificială I delivered the "AI – Artificial Intelligence Workshop" organized by the Chamber of Commerce, Industry and Agriculture Arad. During this workshop, I introduced the basic concepts of generative artificial intelligence, explaining to participants the advantages of this new technology and how they can use it.

[01/04/2023 – 27/04/2023] Arad

Course - Competențe Digitale Trainer for the Digital Skills Course organized by the Chamber of Commerce, Industry and Agriculture Arad. During this course:

- We covered a brief history of the IT field
- We covered basic IT concepts
- We covered the applications offered by the Office suite, focusing on Excel
- Practical exercises with the students

The participants were between 20 and 50 years old.

PROJECTS

[27/10/2025 – 07/11/2025]

DiBaX - NATO Innovation Event Defense industry demonstration project. Successfully demonstrated GenAI application integration with a COP (Common Operational Picture) server to provide real-time, dynamic battlefield data for operator decision-making. Implemented agentic workflow architecture where GenAI interprets user messages, extracts relevant information, and orchestrates multiple agents to achieve desired outcomes.

Technologies: Python, MCP (Model Context Protocol), Elastic Search, Kibana, GenAI orchestration, Prompt Engineering

Link: https://www.linkedin.com/posts/lmtinnovations_dibax-2025-activity-7354051141673922561-uDBn/

[04/02/2025 – 05/02/2025]

Hackathon - PoC: Aria Agent I participated in a hackathon organized by one of the company's major clients, focused on integrating AI agents into enterprise projects and developing new business concepts based on AI technologies.

Duration: 36 hours.

Key contributions:

- Research, design, and rapid development of an orchestrated AI agent prototype (Python, LlamaIndex, custom SignalR-like protocol) integrated with the main .NET application
- Implementation of TTS (Kokoro) and STT (Faster Whisper) integration for bidirectional voice communication with the application
- Orchestration and close collaboration with the team:
 - Integration of database query functionalities and agentic function calls
 - Solving backend–frontend communication issues
- Video directing and scripting: Designed, filmed, and coordinated the creation of the presentation video, managing the script, video material selection, voice-over, and integration of real usage examples
- Multidisciplinary collaboration with colleagues from frontend and marketing departments (international)
- Result: Final presentation of the marketing video for the client + functional demo (worked about 10% of the time)

Technologies: Python, .NET, LlamaIndex, SignalR, TTS Kokoro, Faster Whisper STT, AI agent orchestration, backend-frontend integration, video scripting & directing, enterprise software.

[01/07/2024 – 05/08/2024]

Mentor – PoC: AI Voice Control Agent I designed and coordinated a proof-of-concept project together with an intern, aiming to demonstrate voice control over a SPA web application (.NET) using artificial intelligence.

Project structure:

- STT Server: voice-to-text conversion with Faster Whisper
- AI Server: multi-layer agentic processing, prompt interpretation, and generating function calls to the SPA
- SPA App (.NET): performing CRUD operations via voice command through API endpoints

I guided and assisted the intern throughout all stages (architectural design, troubleshooting, workflow optimization, testing local STT/AI models, performance analysis). Direct coding interventions were minimal, with the focus on skill development, autonomy, and structuring agentic logic.

Result: A functional PoC, with a comparative report between different local AI/STT models, presented in tabular (Excel) format.

Skills gained from this experience:

- Improved technical mentorship and coaching
- Improved leadership and project coordination
- Architectural design of distributed AI systems
- Workflow optimization and local model performance analysis
- Agentic logic thinking and prompt engineering

[01/02/2024 – 15/03/2024]

PoC: AI - local AI connected to local database This was a pilot, proof-of-concept project in which I tried to connect local AI to our internal database and process its information to generate a new dataset. For this project, I dedicated a few hours per week to research & development.

Along the way, I encountered a number of practical issues:

- Our version of EntityFramework cannot be used for reflection with IronPython, so direct integration of C# .dlls into Python code is not possible.
- Our database is extremely complex and full of abbreviations; I had to provide additional context to the AI models to enable them to create and run queries correctly.
- I had to analyze and explain the entire database structure to make it "AI-friendly."
- Local models require a lot of CPU processing time to interact efficiently with the database, which limits their use in real workflows.
- I researched and tested relevant frameworks for AI-database integration: LlamaIndex, LangChain, SQLAlchemy.

Result: A PowerPoint presentation for our clients (actually for the shareholders), outlining what would be required if we decide to implement such a capability in our software in the future.

[01/07/2023 – 27/09/2023]

Mentor - PoC: AI - for Documentation During this period, I had two interns at the company, in consecutive intervals (July 1 - August 5; August 20 - September 27). I've designed a proof-of-concept project using local AI models (Large Language Models) to analyze documents and generate relevant answers based on the contained information, and they had to implement it. Essentially, we developed a local RAG (Retrieval-Augmented Generation) application. My direct intervention was quite minimal, I coordinated, answered questions, and guided their steps, but the actual code was mainly written by the interns. An interesting aspect was the difficulty encountered by the second intern: major compatibility issues between different versions of AI libraries and Python, which can change from one month to the next, leading to additional troubleshooting sessions.

Skills gained from this experience:

- Leadership & technical coordination
- Mentorship and training in local AI (LLM, RAG)
- Improved organization and management of PoC projects in a small team

[01/06/2019 – 15/01/2020]

PoC: Automatic Transportation During this period, together with a colleague, we developed an intermediary program using C# that allowed the production robots to communicate with conveyor-type equipment. A production machine was designed with a conveyor that transported parts inside the equipment. When the machine had no parts to process, a signal needed to be sent to the production robots to place parts onto the conveyor. If the parts had finished processing, a signal had to be sent to notify the robots to come and collect the parts from the conveyor. We worked on the communication between the machine's conveyor and sending messages to the robots, thus creating an intermediary program that enabled communication between these two separate entities. From the robotics team, a colleague handled the robot pathing and the transmission of messages from the robots to our program. Result: by the end of the project, the robots were able to pick up and place parts on the conveyor as needed.

PUBLICATIONS

[2026]

Prevalence and Risk Factors of *Borrelia burgdorferi* Senu Lato IgG Antibodies Among Blood Donors in Western Romania Contributions:

- Data cleaning, analysis, statistical analysis, and technical support

Authors: Alin Gabriel Mihu, Maria Daniela Mot, Daniela Adriana Oatis, Sergiu Adrian Sprintar, Liana Maria Chicea, Rodica Lighezan, Ana Alexandra Ardelean, Maria Alina Lupu, Tudor Rares Olariu | **Journal Name:** Pathogens

[2025]

Seroepidemiology of Severe Acute Respiratory Syndrome Coronavirus 2 Infection in Blood Donors from Western Romania, August-September 2023 Contributions:

- Data cleaning, analysis, statistical analysis, and technical support

Authors: Tudor Rares Olariu, Rodica Lighezan, Sorin Ursoniu, Alina Cristiana Craciun, Alexander Tudor Olariu, Sergiu Adrian Sprintar, Daniela Adriana Oatis, Maria Alina Lupu, Alin Gabriel Mihu | **Journal Name:** Microorganisms

[2025]

First Report on the Seroprevalence and Risk Factors Associated with *Toxocara* Infection in Blood Donors from Romania Contributions:

- Data cleaning, analysis, statistical analysis, and technical support

Authors: Ana Alexandra Ardelean, Rodica Lighezan, Sorin Ursoniu, Sergiu Adrian Sprintar, Daniela Adriana Oatis, Alin Gabriel Mihu, Maria Alina Lupu, Tudor Rareş Olariu | **Journal Name:** Pathogens

[2025]

***Toxoplasma gondii* Seroprevalence and Associated Risk Factors in Psychiatric Patients Diagnosed with Moderate and Major Depression from Western Romania: A Case-Control Retrospective Study** Contributions:

- Data cleaning, analysis, statistical analysis, and technical support

Authors: Alin Gabriel Mihu, Alexander Tudor Olariu, Ligia Elisaveta Piroș, Sebastian Grada, Ana Alexandra Ardelean, Sergiu Adrian Sprintar, Daniela Adriana Oatis, Rodica Lighezan, Tudor Rares Olariu | **Journal Name:** Life

[2025]

Trends in Positive Urine Culture Rates and Antimicrobial Resistance in Non-Hospitalized Children from Western Romania: A Retrospective Observational Study Contributions:

- Data cleaning, analysis, statistical analysis, and technical support

Authors: Constantin Catalin Marc, Maria Daniela Mot, Monica Licker, Delia Muntean, Daniela Teodora Marti, Ana Alexandra Ardelean, Alina Ciceu, Sergiu Adrian Sprintar, Daniela Adriana Oatis, Alin Gabriel Mihu, Tudor Rares Olariu
| **Journal Name:** Antibiotics

[2025]
Retrospective Serologic Assessment of Vitamin D Levels in Children from Western Romania: A Cross-Sectional Study Contributions:
• Data cleaning, analysis, statistical analysis, and technical support

Authors: Alin Gabriel Mihu, Cristian Mircea Nicolescu, Constantin Catalin Marc, Casiana Boru, Monica Susan, Alina Ciceu, Sergiu Adrian Sprintar, Alexander Tudor Olariu, Daniela Adriana Oatis, Laura Corina Nicolescu, Tudor Rares Olariu
| **Journal Name:** Medicina

[2024]
Prevalence and Antibiotic Resistance of Streptococcus agalactiae in Women of Childbearing Age Presenting Urinary Tract Infections from Western Romania Contributions:
• Data cleaning, analysis, statistical analysis, and technical support

Authors: Constantin Catalin Marc, Monica Susan, Sergiu Adrian Sprintar, Monica Licker, Daniela Adriana Oatis, Daniela Teodora Marti, Razvan Susan, Laura Corina Nicolescu, Alin Gabriel Mihu, Tudor Rares Olariu, Delia Muntean
| **Journal Name:** Life

VOLUNTEERING

[01/01/2025 – Current] Timisoara
Researcher
Collaborator with the Center for Diagnosis and Study of Parasitic Diseases, Department of Infectious Diseases, Victor Babeş University of Medicine and Pharmacy Timișoara, supporting medical research projects.

[01/06/2024 – Current] Arad
Researcher
Collaborator with the “Aurel Ardelean” Institute of Life Sciences, Vasile Goldiș Western University of Arad, supporting medical research projects.